RESEARCH IN COMPUTING

PRACTICAL VIVA QUESTIONS & ANSWERS

Complete, sequence-wise revision guide
Practical 1 to Practical 8(b)

M.Sc. DATA SCIENCE | PART I | SEMESTER I

Based on the submitted RIC Practical Journal
Academic Year 2026–2027

Prepared for practical examination revision

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All topics follow the order in the practical journal. Journal page references help you find the original screenshots, datasets, and outputs.

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How to use this guide

For every practical, remember its **Aim**, key **Definition**, **Steps in Excel**, and **Conclusion**. Practice saying each answer in your own words.

Exam tip

For hypothesis testing, remember: reject H_0 if p-value < α; otherwise fail to reject H_0.

Source note: This guide follows the 40-page RIC Practical Journal and reproduces the questions and answers prepared from it in this conversation. The journal may contain some numerical or decision-rule inconsistencies; verify numerical results against your Excel output.

PRACTICAL 1

Descriptive Statistics Using Excel

RIC Journal pages: 1–4 | 12 viva questions

Q1. What is descriptive statistics?

Answer: Descriptive statistics is a method used to summarize, organize, and describe data using measures such as mean, median, mode, standard deviation, minimum, and maximum.

Q2. What is the aim of Practical 1?

Answer: The aim is to obtain descriptive statistics of given data using Microsoft Excel.

Q3. What is mean?

Answer: Mean is the average of all values. It is calculated by dividing the sum of all observations by the total number of observations. Formula: Mean = Sum of values / Number of values.

Q4. What is median?

Answer: Median is the middle value of the data when arranged in ascending or descending order.

Q5. What is mode?

Answer: Mode is the value that occurs most frequently in a dataset.

Q6. What is standard deviation?

Answer: Standard deviation measures how much the values are spread around the mean. A higher standard deviation indicates greater variation.

Q7. What is variance?

Answer: Variance measures the average squared deviation of data values from their mean.

Q8. What is range?

Answer: Range is the difference between the maximum and minimum values. Formula: Range = Maximum − Minimum.

Q9. What is the Analysis ToolPak in Excel?

Answer: Analysis ToolPak is an Excel add-in used to perform statistical analysis, including descriptive statistics, t-tests, ANOVA, and sampling.

Q10. How do you perform descriptive statistics in Excel?

Answer: Go to Data → Data Analysis → Descriptive Statistics → Select Input Range → Tick Summary Statistics → Click OK.

Q11. What is skewness?

Answer: Skewness measures the asymmetry of a data distribution.

Q12. What is kurtosis?

Answer: Kurtosis describes the heaviness of the tails of a data distribution.

PRACTICAL 2

Import Data from Excel and CSV

RIC Journal pages: 5–7 | 10 viva questions

Q1. What is the aim of Practical 2?

Answer: The aim is to import data from Excel and CSV files into Microsoft Excel for further analysis.

Q2. What is a CSV file?

Answer: CSV stands for Comma-Separated Values. It is a simple text file used to store tabular data, usually with values separated by commas.

Q3. What is the difference between Excel and CSV?

Answer: Excel files can contain multiple worksheets, formulas, charts, and formatting, whereas CSV files mainly store plain tabular data.

Q4. What is data importing?

Answer: Data importing is the process of bringing data from an external file or source into an application for processing and analysis.

Q5. How do you import CSV data into Excel?

Answer: Go to Data → Get Data → From File → From Text/CSV → Select File → Load.

Q6. What is the extension of an Excel file?

Answer: The commonly used Excel file extension is .xlsx.

Q7. Why do we import data?

Answer: We import data to organize, clean, process, and analyze information without manually entering every value.

Q8. What is Power Query?

Answer: Power Query is an Excel feature used to import, clean, transform, and combine data from different sources.

Q9. What is a delimiter?

Answer: A delimiter is a character, such as a comma or semicolon, used to separate values in a text file.

Q10. What is the output of Practical 2?

Answer: The selected Excel or CSV dataset is successfully imported and displayed in an Excel worksheet.

PRACTICAL 3(a)

Primary Data Collection and Survey Analysis

RIC Journal pages: 8–13 | 13 viva questions

Q1. What is primary data?

Answer: Primary data is original data collected directly by the researcher through surveys, interviews, questionnaires, or observations.

Q2. What is the aim of Practical 3(a)?

Answer: The aim is to design a survey form, collect primary data, and analyze users' preferences and experiences with an online payment application.

Q3. What is a survey?

Answer: A survey is a method of collecting information from people by asking a set of questions.

Q4. What is a questionnaire?

Answer: A questionnaire is a structured set of questions prepared to collect information from respondents.

Q5. Which case study is used in your practical?

Answer: My practical uses an online payment application survey to understand customer needs, payment preferences, security concerns, and user satisfaction.

Q6. What is a PivotTable?

Answer: A PivotTable is an Excel tool used to summarize, group, compare, and analyze large amounts of data.

Q7. What is frequency distribution?

Answer: Frequency distribution shows how often each value or category appears in a dataset.

Q8. How do you create a PivotTable in Excel?

Answer: Select the dataset → Insert → PivotTable → Select Table/Range → Click OK → Choose the required fields.

Q9. What is a scatter plot?

Answer: A scatter plot is a graph that displays the relationship between two numerical variables using points.

Q10. What is user-friendliness?

Answer: User-friendliness means how easily users can understand and operate an application.

Q11. What are the different online payment methods in your survey?

Answer: The methods include Credit/Debit Card, Mobile Wallet, Net Banking, and UPI.

Q12. What analysis did you perform in Practical 3(a)?

Answer: I performed frequency distribution analysis, PivotTable analysis, and an analysis of usage patterns in relation to user-friendliness.

Q13. What is the result of this practical?

Answer: The analysis helps identify customer preferences, security concerns, satisfaction, and possible improvements for the online payment application.

PRACTICAL 3(b)

Secondary Data Analysis

RIC Journal pages: 14–16 | 10 viva questions

Q1. What is secondary data?

Answer: Secondary data is information that has already been collected by another person or organization for an earlier purpose.

Q2. What is the aim of Practical 3(b)?

Answer: The aim is to perform statistical analysis of existing secondary data using Microsoft Excel.

Q3. Give examples of secondary data.

Answer: Government census reports, research papers, websites, books, and published statistical reports.

Q4. What is the difference between primary and secondary data?

Answer: Primary data is collected directly by the researcher, whereas secondary data is collected from already available sources.

Q5. Which dataset is used in your practical?

Answer: My practical uses World Population Census Data for population analysis and decision-making.

Q6. What is a population pyramid?

Answer: A population pyramid is a graphical representation of population distribution according to age groups and gender.

Q7. Which chart is used for population analysis?

Answer: A horizontal bar chart is used to represent the male and female population across different age groups.

Q8. What is the AutoSum function?

Answer: AutoSum is an Excel feature used to calculate the total of selected numerical values automatically.

Q9. What is the purpose of secondary data analysis?

Answer: Its purpose is to identify existing trends, patterns, and relationships without collecting new data.

Q10. What are the steps of secondary data analysis?

Answer: Define the research question, locate data, evaluate its relevance, assess credibility, and perform analysis.

PRACTICAL 4(a)

Two-Sample t-Test

RIC Journal pages: 17–19 | 10 viva questions

Q1. What is hypothesis testing?

Answer: Hypothesis testing is a statistical method used to determine whether sample evidence supports or contradicts a statement about a population.

Q2. What is a t-test?

Answer: A t-test is a statistical test used to compare means and determine whether the observed difference is statistically significant.

Q3. What is a two-sample t-test?

Answer: A two-sample t-test compares the means of two independent groups.

Q4. What is the aim of Practical 4(a)?

Answer: The aim is to test whether there is a significant difference between the mean scores of two groups using a t-test.

Q5. What is a null hypothesis (H_0)?

Answer: The null hypothesis states that there is no statistically significant difference or effect.

Q6. What is an alternative hypothesis (H_1)?

Answer: The alternative hypothesis states that there is a statistically significant difference or effect.

Q7. What is the level of significance?

Answer: The level of significance, denoted by alpha (α), is the threshold used to decide whether to reject the null hypothesis. Common values are 0.05 and 0.01.

Q8. What is a p-value?

Answer: A p-value measures how strongly the sample evidence conflicts with the null hypothesis. If the p-value is less than alpha, we reject H_0.

Q9. How do you perform a t-test in Excel?

Answer: Go to Data → Data Analysis → Select the appropriate t-Test option → Enter both input ranges → Select alpha → Click OK.

Q10. What is the conclusion of your practical?

Answer: The journal concludes that the experimental and comparison groups have a statistically significant difference in their scores.

PRACTICAL 4(b)

Paired t-Test

RIC Journal pages: 20–22 | 9 viva questions

Q1. What is a paired t-test?

Answer: A paired t-test compares the means of two related sets of observations, such as measurements before and after treatment.

Q2. What is the aim of Practical 4(b)?

Answer: The aim is to test the difference between paired observations using a paired t-test.

Q3. What is the difference between a paired and an independent t-test?

Answer: A paired t-test is used for related observations, while an independent t-test is used for two unrelated groups.

Q4. Give an example of a paired t-test.

Answer: Comparing the same students' marks before and after a training program.

Q5. Which Excel tool is used for a paired t-test?

Answer: Data → Data Analysis → t-Test: Paired Two Sample for Means.

Q6. What is the paired t-test formula?

Answer: t = (mean difference) / (standard deviation of differences / √n). Here, n is the number of pairs.

Q7. What is the calculated t-value in your practical?

Answer: The calculated t-value is approximately 2.9963.

Q8. What is the critical t-value in your practical?

Answer: The critical two-tailed t-value is approximately 2.8609 at the 1% significance level.

Q9. What is the conclusion of Practical 4(b)?

Answer: Since the calculated t-value is greater than the critical t-value, the null hypothesis is rejected. The difference is statistically significant.

PRACTICAL 5(a)

Chi-Square Goodness-of-Fit Test

RIC Journal pages: 23–24 | 10 viva questions

Q1. What is a Chi-Square test?

Answer: A Chi-Square test is a statistical test used to analyze categorical data by comparing observed and expected frequencies.

Q2. What is the Chi-Square goodness-of-fit test?

Answer: It determines whether observed frequencies follow a specified expected distribution.

Q3. What is the aim of Practical 5(a)?

Answer: The aim is to check whether preferences for different computer operating systems follow an equal distribution.

Q4. Which operating systems are used in your practical?

Answer: Windows, Mac, and Linux.

Q5. What is observed frequency?

Answer: Observed frequency is the actual number of occurrences recorded in the data.

Q6. What is expected frequency?

Answer: Expected frequency is the number of occurrences predicted under the null hypothesis.

Q7. What is the Chi-Square formula?

Answer: $\chi^2 = \Sigma (O - E)^2 / E$, where O represents observed frequency and E represents expected frequency.

Q8. What is the degree of freedom in goodness-of-fit?

Answer: For categories with fully specified expected probabilities, degrees of freedom equal the number of categories minus one.

Q9. What is the null hypothesis in your practical?

Answer: All three operating systems are equally preferred by the population.

Q10. What is the conclusion of your practical?

Answer: The journal reports a Chi-Square value of approximately 3.20 and a critical value of approximately 5.99. Since the calculated value is smaller, we fail to reject the null hypothesis.

PRACTICAL 5(b)

Chi-Square Test of Independence

RIC Journal pages: 25–27 | 10 viva questions

Q1. What is a Chi-Square test of independence?

Answer: It is used to determine whether two categorical variables are statistically associated.

Q2. What is the aim of Practical 5(b)?

Answer: The aim is to examine whether students' grades are associated with gender.

Q3. What is a contingency table?

Answer: A contingency table displays the frequencies of two categorical variables in rows and columns.

Q4. What is the null hypothesis?

Answer: The students' grade distribution is independent of gender.

Q5. What is the alternative hypothesis?

Answer: The students' grade distribution is associated with gender.

Q6. How do you calculate expected frequency?

Answer: Expected Frequency = (Row Total × Column Total) / Grand Total.

Q7. What is the degree of freedom formula?

Answer: df = $(r - 1)(c - 1)$, where r is the number of rows and c is the number of columns.

Q8. Which Excel function counts specific grades?

Answer: The COUNTIF() function is used to count the number of students receiving a particular grade.

Q9. Which function calculates the Chi-Square critical value?

Answer: =CHISQ.INV.RT(0.05,4) calculates the critical value at the 5% level with four degrees of freedom.

Q10. What is the conclusion of Practical 5(b)?

Answer: The journal's calculated Chi-Square value is approximately 1.235 and the critical value is approximately 9.488. Since the calculated value is smaller, we fail to reject H_0.

PRACTICAL 6

Z-Test

RIC Journal pages: 28–29 | 10 viva questions

Q1. What is a Z-test?

Answer: A Z-test is a statistical hypothesis test that uses the standard normal distribution to test means or differences between means under appropriate conditions.

Q2. What is the aim of Practical 6?

Answer: The aim is to perform hypothesis testing using a two-sample Z-test in Excel.

Q3. When is a Z-test used?

Answer: A Z-test is commonly used when observations are independent and the population standard deviation is known, or when suitable large-sample approximations apply.

Q4. What is the difference between a Z-test and a t-test?

Answer: A Z-test uses the standard normal distribution, while a t-test uses the t-distribution and is commonly used when the population standard deviation is unknown.

Q5. What is the Z-test formula for one population mean?

Answer: Z = (sample mean − population mean) / (population standard deviation / √n).

Q6. What is the null hypothesis in your practical?

Answer: The population means of the two groups are equal, meaning their difference is zero.

Q7. What is the alternative hypothesis?

Answer: The population means of the two groups are different.

Q8. How do you perform a Z-test in Excel?

Answer: Go to Data → Data Analysis → z-Test: Two Sample for Means → Enter data ranges and known variances → Click OK.

Q9. What is a two-tailed test?

Answer: A two-tailed test checks whether a population parameter differs from its hypothesized value in either direction.

Q10. What is the result of your practical?

Answer: The journal's output rejects the null hypothesis, indicating a statistically significant difference between the two means.

PRACTICAL 7(a)

One-Way ANOVA

RIC Journal pages: 30–31 | 10 viva questions

Q1. What is ANOVA?

Answer: ANOVA stands for Analysis of Variance. It is a statistical method used to compare the means of multiple groups.

Q2. What is one-way ANOVA?

Answer: One-way ANOVA compares the means of three or more groups based on one independent factor.

Q3. What is the aim of Practical 7(a)?

Answer: The aim is to compare the mean scores of different groups using one-way ANOVA in Excel.

Q4. Why do we use ANOVA instead of multiple t-tests?

Answer: ANOVA compares multiple group means in one overall test and helps avoid increasing the risk of Type I errors caused by performing many separate t-tests.

Q5. What is the null hypothesis in ANOVA?

Answer: All population group means are equal.

Q6. What is the alternative hypothesis?

Answer: At least one population group mean is different.

Q7. What is the F-value?

Answer: The F-value is the ratio of variation between groups to variation within groups, adjusted by their degrees of freedom.

Q8. What are the assumptions of one-way ANOVA?

Answer: Independent observations, approximately normal errors within groups, and approximately equal population variances.

Q9. How do you perform one-way ANOVA in Excel?

Answer: Data → Data Analysis → ANOVA: Single Factor → Select Input Range → Set Alpha → Click OK.

Q10. What is the conclusion of ANOVA?

Answer: If the p-value is less than the significance level, we reject H_0 and conclude that at least one group mean differs.

PRACTICAL 7(b)

Two-Way ANOVA

RIC Journal pages: 32–33 | 10 viva questions

Q1. What is two-way ANOVA?

Answer: Two-way ANOVA is a statistical test used to examine the effects of two independent factors on a continuous dependent variable.

Q2. What is the aim of Practical 7(b)?

Answer: The aim is to perform a two-way ANOVA using Microsoft Excel.

Q3. What is the difference between one-way and two-way ANOVA?

Answer: One-way ANOVA studies one independent factor, whereas two-way ANOVA studies two independent factors.

Q4. What is a factor in ANOVA?

Answer: A factor is an independent categorical variable that may influence the dependent variable.

Q5. What is an interaction effect?

Answer: An interaction effect occurs when the effect of one factor depends on the level of another factor.

Q6. What is replication in ANOVA?

Answer: Replication means having multiple observations for each combination of factor levels.

Q7. Which Excel option is used in your practical?

Answer: ANOVA: Two-Factor With Replication.

Q8. How do you perform two-way ANOVA in Excel?

Answer: Go to Data → Data Analysis → ANOVA: Two-Factor With Replication → Enter Input Range, Rows per Sample, Alpha, and Output Range → Click OK.

Q9. What is the usual significance level?

Answer: The commonly used significance level is 0.05, or 5%.

Q10. What is the conclusion of Practical 7(b)?

Answer: The journal reports rejection of the null hypothesis in its two-way ANOVA output. The test is used to examine significant factor and interaction effects.

PRACTICAL 8(a)

Random Sampling

RIC Journal pages: 34–35 | 10 viva questions

Q1. What is sampling?

Answer: Sampling is the process of selecting a smaller group from a larger population for statistical analysis.

Q2. What is random sampling?

Answer: Random sampling is a method in which observations are selected by chance. In simple random sampling, each population member has an equal probability of selection.

Q3. What is the aim of Practical 8(a)?

Answer: The aim is to select and analyze a random sample of student marks using Excel.

Q4. What is population?

Answer: Population is the entire group of individuals or observations being studied.

Q5. What is a sample?

Answer: A sample is a smaller subset selected from a population.

Q6. What is sample size?

Answer: Sample size is the number of observations selected for analysis.

Q7. How many students are included in your practical?

Answer: The population contains 100 students, and a random sample of 10 marks is selected.

Q8. Which Excel tool is used for random sampling?

Answer: Data → Data Analysis → Sampling.

Q9. How do you perform random sampling in Excel?

Answer: Select the Input Range → Choose Random → Enter Number of Samples as 10 → Select Output Range → Click OK.

Q10. What are the advantages of random sampling?

Answer: Random sampling reduces systematic selection bias and can help obtain a representative sample of the population.

PRACTICAL 8(b)

Periodic Sampling

RIC Journal pages: 36–37 | 10 viva questions

Q1. What is periodic sampling?

Answer: Periodic sampling is a method of selecting observations at a fixed interval from an ordered dataset.

Q2. What is another name for periodic sampling?

Answer: It is closely related to systematic sampling.

Q3. What is the aim of Practical 8(b)?

Answer: The aim is to select and analyze student marks using periodic sampling in Microsoft Excel.

Q4. What is sampling interval?

Answer: Sampling interval is the fixed gap between selected observations.

Q5. What is the sampling interval formula?

Answer: k = N / n, where N is the population size and n is the required sample size.

Q6. What is the sampling interval in your practical?

Answer: The population contains 100 students, and the sample size is 10. Therefore, k = 100 / 10 = 10.

Q7. How do you perform periodic sampling in Excel?

Answer: Data → Data Analysis → Sampling → Select Input Range → Choose Periodic → Enter Period as 10 → Click OK.

Q8. What is the difference between random and periodic sampling?

Answer: Random sampling selects observations by chance, while periodic sampling selects observations at fixed intervals.

Q9. What is the advantage of periodic sampling?

Answer: It is simple, quick, and useful for selecting observations from a large ordered dataset.

Q10. What is a limitation of periodic sampling?

Answer: It may introduce bias if the data contains a repeating pattern that matches the selected interval.

EXAM REVISION

Most Important Common Viva Questions

Short answers for fast revision before the practical examination.

Q1. What is RIC?

Answer: Research in Computing.

Q2. What is research?

Answer: A systematic process of investigating a problem and finding answers.

Q3. What is data analysis?

Answer: The process of examining data to identify useful information.

Q4. What is Excel?

Answer: A spreadsheet application for data storage, calculation, and analysis.

Q5. What is a hypothesis?

Answer: A testable statement about a population or relationship.

Q6. What is H_0?

Answer: Null hypothesis.

Q7. What is H_1?

Answer: Alternative hypothesis.

Q8. What is alpha?

Answer: The significance level used in a hypothesis test.

Q9. What is a Type I error?

Answer: Rejecting a true null hypothesis.

Q10. What is a Type II error?

Answer: Failing to reject a false null hypothesis.

Q11. What is a statistical test?

Answer: A method used to evaluate a hypothesis using sample data.

Q12. What is correlation?

Answer: A measure of the relationship between two variables.

Q13. What is statistical significance?

Answer: Evidence against H_0 strong enough to reject it at the selected alpha level.

Q14. What is the difference between a parameter and a statistic?

Answer: A parameter describes a population; a statistic describes a sample.

Q15. What is data visualization?

Answer: Representing information using charts and graphs.